

Reviewer Response: Local Smoothing and Extrapolation

Super Writer worked example

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Educational mock response. These comments were written for the repository; they did not come from a conference. The response uses only the measurements in the companion toy regression example. This generic layout is not a venue’s official rebuttal template.

R1: The robustness claim is broader than the evidence

Comment. The draft claims robustness to distribution shift, although the extrapolation results appear worse for five neighbors. Please reconcile the abstract with the table.

Response. We agree. The in-domain mean MSE decreases from 0.1083 to 0.0851, but the extrapolation mean increases from 2.6891 to 3.7012. We removed the general robustness claim. The revised abstract reports both domains, and the Results section states that the in-domain advantage does not extend to this tested extrapolation setting. The Interpretation section explains why nearest-neighbor predictions are constant beyond the rightmost training input, without claiming that this determines the ranking on every function.

R2: The uncertainty notation is unclear

Comment. Are the error bars confidence intervals? Were grid points treated as independent experimental runs?

Response. The table reports sample standard deviation across five training seeds, not a confidence interval. Each seed produces one MSE per method and domain by averaging over 101 fixed grid points. The grid points are not counted as separate training replications. We now state the aggregation unit in the Experimental Protocol and label the column “Sample SD.” We also removed “significantly” from the original claim because no significance test was run.

R3: Add a parametric baseline

Comment. A parametric model could be a stronger extrapolation baseline. Can the paper establish superiority over that alternative?

Response. The supplied experiment does not include that comparison, so it cannot establish such superiority. We identify the missing baseline explicitly in Limitations and retain the narrower comparison between one and five neighbors. We have not added a result, promised a completed run, or represented the suggested experiment as evidence already available.

Revision Locations

Comment	Change	Location in the companion manuscript
R1	Report both directions; narrow robustness claim	Abstract; Results; Interpretation
R2	Identify seed-level sample SD; remove significance claim	Experimental Protocol; Results table
R3	State unevaluated baseline and limit comparison	Limitations